12/10/26

Signal Processing

- Recap: Z-transform properties – ROC is important to keep in mind.
- Preparation for LTI system analysis
  - LCCDE motivation: IIR, filter design
  - LCCDE definition and properties
  - Moving average example
- Recap: $\underbrace{\text{Delay}}_{R_x \pm \text{zeros or poles at } 0, \infty}$, $\underbrace{\text{multiplication by } z_0^n}_{|z_0| \cdot R_x}$, $\underbrace{\text{convolution}}_{R_{x_1} \cap R_{x_2}}$. Please review other props in Chap 3.
- $T\{\}$: map from an input seq $x[n]$ to a desired output seq. $y[n]$
- LTI systems:
  - Satisfies both linearity and time invariance properties.
  - $h[n] = T\{\delta[n]\}$ uniquely determines the system
  - $y[n] = h[n] * x[n]$, $*$ is the convolution operator

    $$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

  - $\underbrace{*}_{\text{conv}} \xleftrightarrow{\text{DTFT}} \underbrace{\cdot}_{\text{Multiplication}}$

    $$\Rightarrow Y(e^{j\omega}) = H(e^{j\omega}) \cdot X(e^{j\omega}) \quad \text{or} \quad H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} \quad (X(e^{j\omega}) \neq 0)$$

- We will work with Linear Constant Coefficient Difference Equations (LCCDEs) in our analysis of LTI systems.
- Why?

1: $H(z) = \frac{z - 1/2}{z - 3/2} = \frac{1 - 1/2\, z^{-1}}{1 - 3/2\, z^{-1}} = \frac{Y(z)}{X(z)}$; $|z| < \frac{3}{2}$

$$\underset{\sim}{Y(z)}\left(1 - \frac{3}{2}\underset{\sim}{z^{-1}}\right) = X(z)\left(1 - \frac{1}{2}z^{-1}\right) \quad - ① \quad ; \quad \underbrace{h[n] = \frac{1}{3}\delta[n] - \left(\frac{3}{2}\right)^{n-1} u[-n-1]}_{\text{Implementation in timedom is not practical since we have an infinite impulse response.}}$$

$$y[n] - \frac{3}{2}y[n-1] = x[n] - \frac{1}{2}x[n-1]$$

$$y[n] = x[n] - \frac{1}{2}\underbrace{x[n-1]}_{\text{prev input}} + \frac{3}{2}\underbrace{y[n-1]}_{\text{prev output}} \quad - ②$$

- Efficient implementation of the same system. since we need only two delay elements.

- Needs initial conditions, however.

2: Having poles and zeros allows for control on system design.

3. $h[n] = \frac{1}{M+1}\left(u[n] - u[n-(M+1)]\right) = \frac{1}{M+1}\left(\delta[n] - \delta[n-M]\right) * u[n]$

$$y[n] = \frac{1}{M+1}\sum_{k=0}^{M} x[n-k]$$

Q: Can this system be expressed in the form of a feedback system?

A: Yes. Give it a try.

- The general form of an LCCDE:

$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{m=0}^{M} b_m \cdot x[n-m] \qquad \boxed{y[n] = x[n] + a} \xleftarrow{} \text{Not LTI}$$

Assuming the initial conditions allow for this to be an LTI system, we have

$$\sum_{k=0}^{N} a_k \cdot Y(z) \cdot z^{-k} = \sum_{m=0}^{M} b_m\, X(z) \cdot z^{-m}$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{m=0}^{M} b_m z^{-m}}{\sum_{k=0}^{N} a_k z^{-k}} = \left(\qquad \right) \frac{\prod(\qquad)}{\prod(\qquad)}$$